

Statistics for International Commerce

Week 13: Special Lecture on Modern Analytics Workflow in Business and Research

How Real Data Projects Are Built Today

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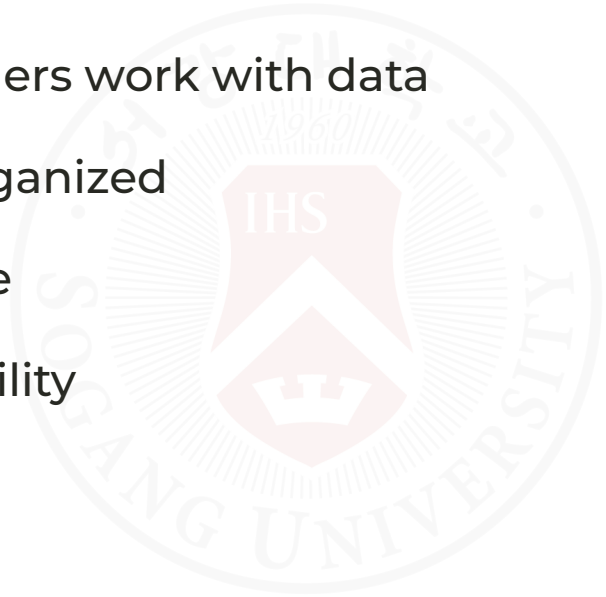
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Today

How Modern Analytics Actually Works

We will talk about:

- how companies and researchers work with data
- how analytics projects are organized
- modern tools used in practice
- collaboration and reproducibility
- AI-assisted workflows

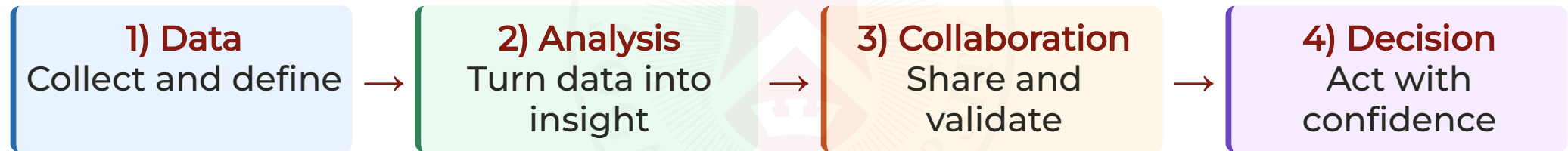


Big Idea

Modern analytics is NOT just:

- statistics
- coding
- charts

It is an entire workflow:



Typical Analytics Workflow



Real-World Example

Suppose a company asks:

“Why are sales falling in Europe?”

A modern analytics workflow may involve:

Pause and discuss: if sales fall in Europe, which part of the workflow would you check first?



Team question:
What would you do in step 1, and what would you show in step 3?

Real tools:
database + notebook + dashboard + collaborative writing

Why This Matters

These workflows are used in:

Ask the room: where do you think analytics is most important in everyday business and research?

Consulting
client decisions

Finance
risk and
forecasting

Fintech
products and users

E-commerce
sales and behavior

Logistics
routes and delivery

Research
evidence and
papers

Central Banks
policy and stability

Tech
products and
analytics

Modern Analytics Is Collaborative

Very few real projects are done alone.

Modern projects involve:

Who does what? Each role sees the project from a different angle.

Analysts: find patterns

Managers: decide priorities

Developers: build tools

Researchers: test ideas

Designers: make it clear

Domain experts: add context

Discussion cue: what goes wrong if one of these roles is missing?

Typical Workflow Tools

Quick question: which of these tools do you already know, and which one would help you most right now?

Coding
Python, R

Development
VS Code

Cloud notebooks
Google Colab

Version control
GitHub

Writing/reporting
Overleaf

Visualization + AI
Tableau, Power BI, Copilot

Step 1. Working with Data

Data can come from:

Company: databases

Web: APIs, websites

Files: Excel, text documents

People: surveys

Markets: financial data

Think first: where would you look if you needed sales data for one country over time?

Example Types of Data

Numerical
sales revenue

Categorical
customer type

Time series
stock prices

Text
reviews, reports

Images
product photos

Example: Python in Analytics

```
##           region      sales quarter
## 0 North America  79.807352      Q3
## 1           Asia  50.828318      Q2
## 2 North America 172.319214      Q3
## 3 North America 156.028602      Q3
## 4           Asia 159.351075      Q3
```

Python helps us:

Organize

clean and structure data

Analyze

find patterns and
relationships

Automate

repeat tasks faster and
safer

Why Python Became So Popular

Class prompt: why do you think Python spread so quickly across business and research?

Easy to start

simple syntax and readable code

Flexible

data, web, automation, AI

Widely used

companies, banks, startups

Community

libraries, tutorials, examples

Why We Use Google Colab

Imagine you are in class: what makes Colab easier for beginners than installing software locally?

No installation

Easy sharing

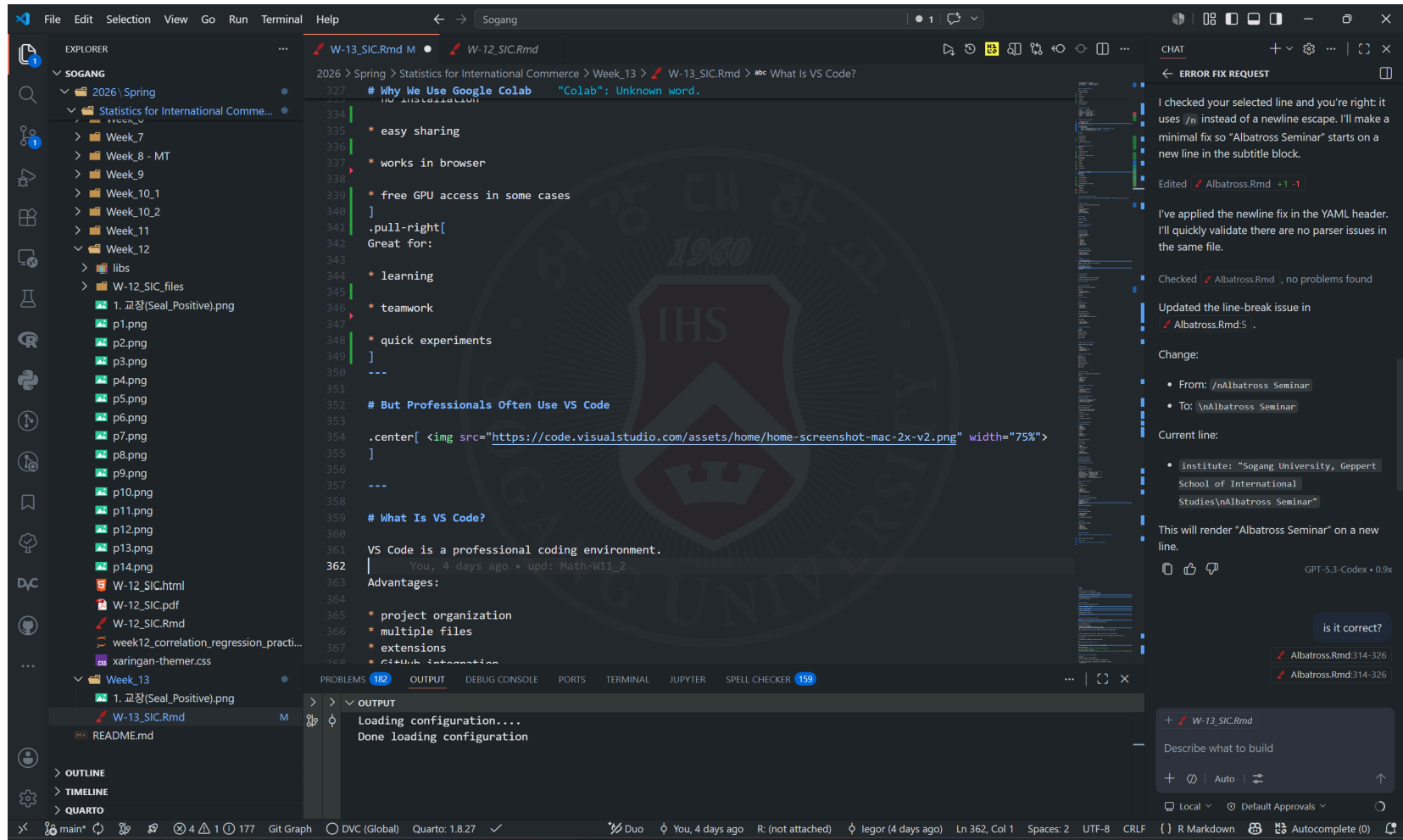
Works in browser

Good for teamwork

Quick experiments



But Professionals Often Use VS Code



What Is VS Code?

VS Code is a professional coding environment.

Advantages:

Why professionals like it: which feature matters most when a project gets bigger?

Project organization

multiple folders and files

Extensions

extra tools for data work

GitHub integration

version control in one place

AI tools

support for coding and debugging

But Remember

The goal is NOT:

“learn software”

The goal is:

solve problems with data

Tools are just tools.

Big idea: the tool matters less than whether it helps you solve the actual problem.

Example Workflow

Try this mentally: if the data are messy, which step should happen before any chart is made?

1. Load

bring data into the project

2. Clean

fix missing values and errors

3. Analyze

find patterns and answer questions

4. Visualize

make the result easy to see

5. Share

present to others

6. Decide

turn insight into action

Step 2. Visualization

Visualization helps people:

Understand
see patterns faster

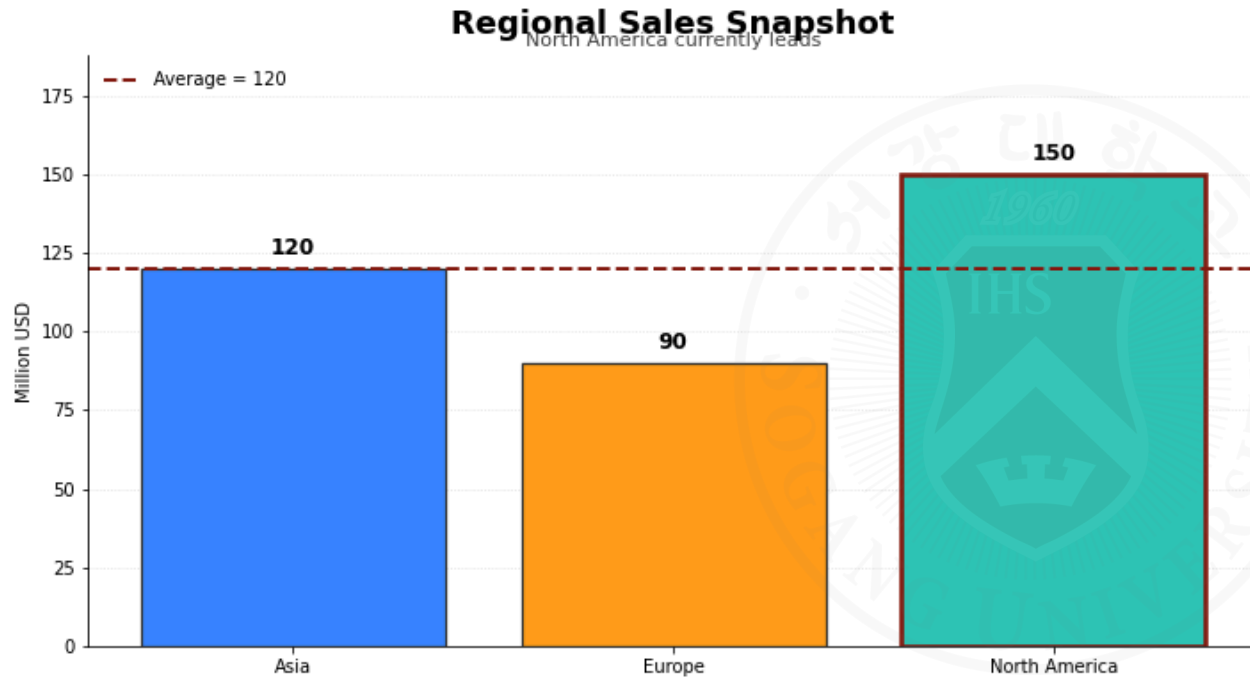
Detect
spot problems early

Communicate
share findings clearly



Example

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Important Reality

In real business:

- communication matters as much as analysis

A brilliant analysis nobody understands:

has little value

Check your message: if you had 30 seconds to explain the result, what would you say first?

Step 3. Collaboration

Modern projects involve many people.

Problem:

- multiple versions
- accidental overwriting
- confusion

Solution:

version control



GitHub

GitHub is used to:

Store code

Track changes

Collaborate

Manage projects



Why GitHub Matters

Without version control:

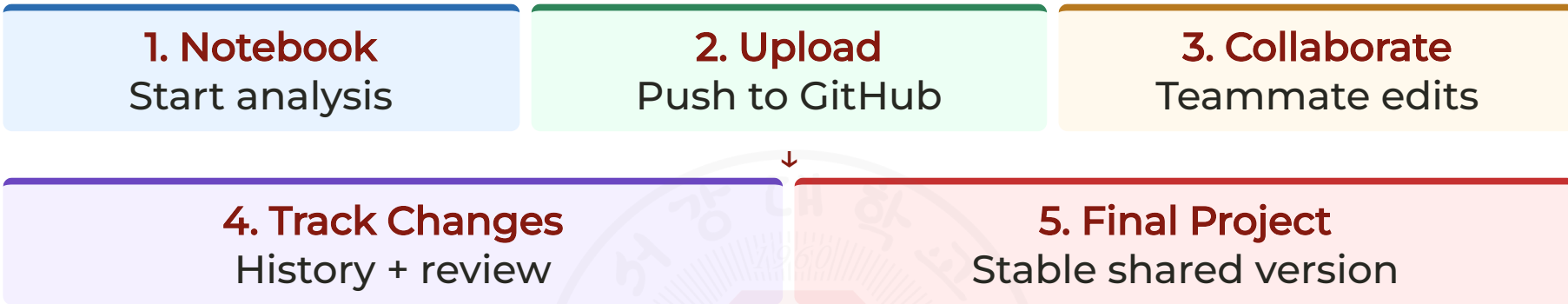
```
text id="f6k3vn" final_project_REAL_final_v2_LAST.xlsx
```

With GitHub:

Without GitHub
file chaos and confusion

With GitHub
history, safety, reproducibility

Example GitHub Workflow



Modern Research Works Similarly

Research today often includes:

Think of a real paper: what do you need besides the text itself to make it convincing?

Code

for cleaning and analysis

Datasets

the evidence base

Figures & Tables

what the reader sees first

Collaborative Writing

drafts, comments, revisions

Reproducibility

someone else should be able to follow your steps

Discussion cue: which part becomes the bottleneck when the project gets bigger?

Example Research Workflow

1. **Data Collection** from surveys, experiments, or databases



2. **Python Analysis** for cleaning, modeling, and testing



3. **Figures & Tables** to communicate core findings



4. **GitHub Collaboration** for versioning and co-author work



5. **Overleaf Paper** for structured writing and references



6. **Publication** as reproducible output

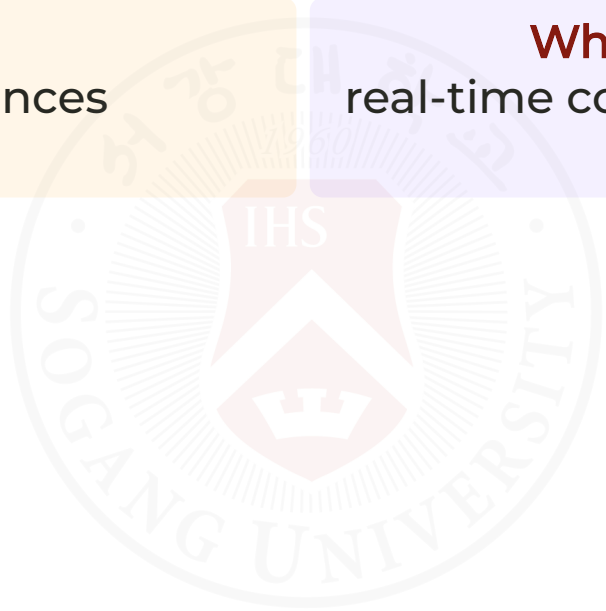
Think like a co-author: what makes writing easier when several people edit the same document?

What it is
a collaborative writing platform

Used for
papers, reports, presentations

Also useful for
equations and references

Why researchers like it
real-time collaboration, version history,
cloud access



Example of Reproducibility

Question: if a classmate runs your code, do they get the same answer?

Reproduce
same result again

Transparent
clear calculations

Documented
workflow is written down

Why Reproducibility Matters

Without reproducibility
harder to find mistakes, weaker trust,
unclear results

With reproducibility
easier review, stronger credibility, safer
decisions

Discussion cue: where would reproducibility matter more, a business report or a research paper?

AI Is Changing Analytics

AI is changing the workflow: where would you let AI help, and where would you still check manually?

Coding

generate or explain code

Debugging

spot errors faster

Summarizing

compress long text or notes

Writing + Visualization

draft text and suggest charts

ChatGPT

GitHub Copilot

Cursor AI

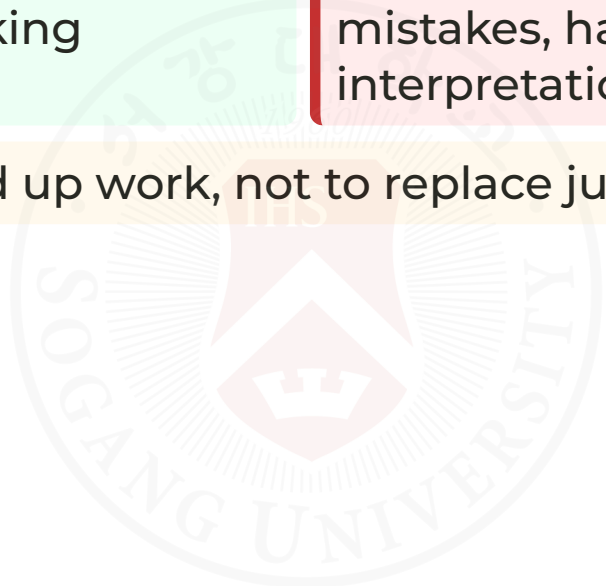
But Important Warning

AI is a tool.

You still need
logic, statistics, critical thinking

AI can produce
mistakes, hallucinations, wrong
interpretation

Rule of thumb: use AI to speed up work, not to replace judgment.



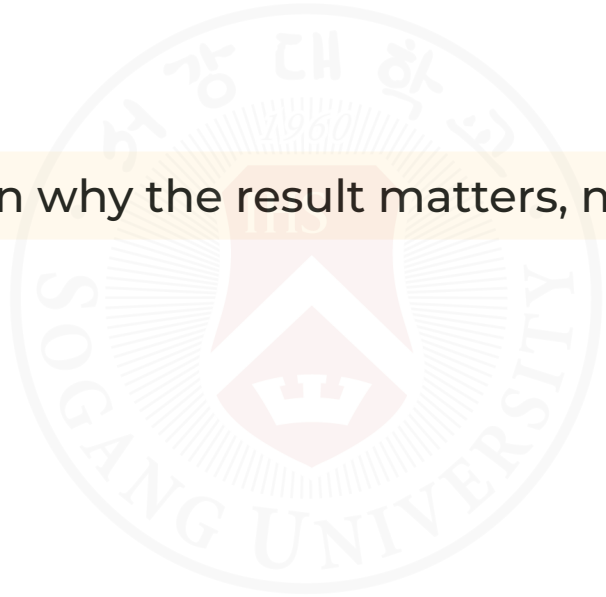
The Most Valuable Skill

Not memorizing formulas.

The most valuable skill is:

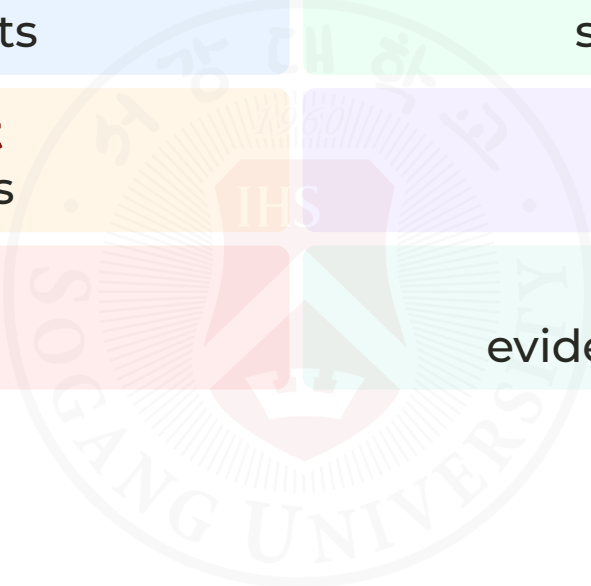
turning data into decisions

Final check: can you explain why the result matters, not just what the result is?



Example Careers

These workflows are used in:



Data analyst dashboards, reports	Business analyst strategy decisions
Research analyst statistical analysis	Financial analyst forecasting
Consultant client analytics	Policy analyst evidence-based decisions

What Employers Often Want

Not only:

- technical skills

But also:



What You Already Know

From this course, you already learned:

- descriptive statistics
- probability
- hypothesis testing
- ANOVA
- Python basics

These are real foundations.

You already have a toolkit: the next step is connecting it to real problems, not starting from zero.

Important Message

You do NOT need to become:

- expert programmer
- mathematician
- AI engineer

to use analytics effectively.

Takeaway: you do not need to be the best coder in the room to make a useful analysis.

Final Idea

Modern analytics combines:

Stats

Coding

Communication

Collaboration

Business



One Sentence Summary

Data is valuable only when it helps people make better decisions.

Final question: what is one decision in your life or work that could be better with data?



The background of the slide features a large, faint, circular watermark of the Seoul National University (SNU) logo. The logo contains the university's name in Korean and English, the year 1960, and a central shield emblem.

Any questions?

Thank you for your attention and active participation!